

Playwright with TypeScript

Week 5 – Home Assignment



Program	Playwright with TypeScript
Module	TypeScript OOP Concepts
Week	Week 5
Assignment	Method Overloading and Method Overriding (Automation Testing Context)

1. Objective

To practice method overloading and method overriding in TypeScript using real-time automation testing scenarios.

2. Problem Statement

Create a TypeScript program to demonstrate method overloading and method overriding using a base class for common Playwright actions and child classes for specific element handling actions.

3. Requirements

- Create a parent class called **BaseActions** with the following:
 - Property: `pageName` (string)
 - Overload the method `click()` with the following signatures:
 - `click(locator: string): void`
 - `click(locator: string, description: string): void`
 - Implement the method to print a meaningful message using `console.log()`.
- Create a child class called **ButtonActions** that extends **BaseActions** with the following:
 - Override the `click(locator: string, description: string)` method to add a custom log message for button clicks and then call the parent method using `super`.
 - Add a new method `clickAndVerifyText(locator: string, expectedText: string)` to simulate clicking and verifying text.
- Create another child class called **InputActions** that extends **BaseActions** with the following:
 - Override the `click(locator: string, description: string)` method to add a custom log message for input clicks and then call the parent method using `super`.
 - Add a new method `fill(locator: string, value: string)` to simulate entering text.
- Create objects for both **ButtonActions** and **InputActions** and call the methods to demonstrate method overloading and overriding.
- Add meaningful comments and use proper TypeScript types.

4. Sample Test Data and Expected Output

Sample Test Data (Code)	Expected Output (Console)
<pre>// ButtonActions object const button = new ButtonActions("Button Page"); button.click("#submitBtn"); button.click("#submitBtn", "Submit Button"); button.clickAndVerifyText("#submitBtn", "Submit"); // InputActions object const input = new InputActions("Input Page"); input.click("#username", "Username Field"); input.fill("#username", "Admin");</pre>	<pre>BaseActions: Clicked on #submitBtn ButtonActions: Clicking on button - Submit Button BaseActions: Clicked on #submitBtn (Submit Button) ButtonActions: Clicked and verifying text 'Submit' on #submitBtn InputActions: Clicking on input field - Username Field BaseActions: Clicked on #username (Username Field) InputActions: Entering text 'Admin' in #username</pre>

5. Technical Guidelines

- Use TypeScript (.ts file).
- Implement method overloading in the parent class.
- Implement method overriding in the child classes using the `override` keyword.
- Use real-time automation testing method names (e.g., `click`, `fill`).
- Use `console.log()` to display the output.
- Add comments for better understanding.
- Follow proper naming conventions and code formatting.

6. Learning Outcome

After completing this assignment, you should be able to:

- Understand and implement method overloading in TypeScript.
- Understand and implement method overriding in TypeScript.
- Apply these concepts using real-time automation testing methods.
- Work with parent and child classes in a practical scenario.